Tactix: Sweep Completion-State Parity Finding

Report to Andrew

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Outcome

A current-source N4 audit isolated one remaining Sweep lifecycle difference. If a resumed tail is entirely filtered at call entry, Kissat and CaDiCaL's schedule/unschedule state machine increments completed passes twice; Tactix increments once. The source-level derivation is documented in the AM425 technical report and the living `SP-SWEEP-3` ledger row. No solver or proof route changed.

Evidence and interpretation

The maintained Tactix retained-tail fixture asserts that the old pure seed is filtered, a separate fresh eligible seed is not admitted in that call, and the completed counter advances once. It passed on current Release and ASan+UBSan binaries. Both pinned references have explicit incomplete flags; their schedule phase recognizes the old pass as complete and their unschedule phase recognizes the new empty pass as complete. This is source inference, not an instrumented reference measurement.

At initial count zero, their next default Sweep call would have limits of 1,024 environment variables and 4,096 clauses, while Tactix would have 512 and 2,048; all three use depth 3 because that cap has already saturated. We have not measured how often this state occurs or whether the faster ramp helps. Copying it now could shift proof-producing environments and cumulative performance. The ledger therefore remains open, not "parity achieved." Documentation option coverage, citations, section references, size, proof-rule sync and rule/policy boundaries all pass.

Next action and gates

Replay the complete lifecycle on a common formula/state with instrumented references, then inspect seed/environment/proof outcomes and a checked cumulative campaign before a general scheduler repair or measured deviation. AM401/412/414 owner jobs remain active; the H2 no-option DSR contract, broad N1/N2, N3 performance, N4 parity, and online-learning production refactors are still open. The requested full 5,000-second Tier-D run remains contingent on those gates and has not been submitted.